

GRANDE PRAIRIE, AB, Canada

WMO: 719400

Lat: 55.180N

Lon: 118.880W

Elev: 2195

StdP: 13.57

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-23.2	-39.1	0.6	-30.8	-31.1	1.0	-23.0	30.1	36.9	28.3	36.5	2.8	330	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	81.4	61.7	78.1	60.5	75.1	59.1	64.5	77.1	62.5	74.4	60.6	72.0	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.9	83.6	69.1	57.8	77.3	66.7	55.8	72.0	64.9	30.7	77.3	29.2	74.3	27.8	72.1	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.1	22.3	19.1	DB	-38.6	87.4	7.8	3.3	-44.2	89.8	-48.8	91.7	-53.2	93.6	-58.8	96.0	(4)
(5)			WB	-38.8	67.8	7.7	2.2	-44.3	69.4	-48.8	70.7	-53.1	72.0	-58.7	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	8.4	12.5	21.5	38.2	50.4	57.7	61.6	59.4	50.9	38.5	21.0	10.4		
	DBStd	22.43	17.79	14.95	14.15	9.15	7.74	5.20	5.02	5.81	7.26	8.46	13.30	15.67		
	HDD50	6218	1290	1051	882	358	89	5	1	4	76	362	871	1229		
	HDD65	10627	1755	1471	1347	803	454	224	128	188	423	820	1320	1694		
	CDD50	1111	0	0	0	6	101	237	361	296	103	7	0	0		
	CDD65	46	0	0	0	0	1	6	23	16	0	0	0	0		
	CDH74	892	0	0	0	2	54	132	359	282	62	1	0	0		
	CDH80	158	0	0	0	0	5	16	71	58	8	0	0	0		
	WSAvg	7.4	6.0	6.0	7.1	8.7	9.3	8.8	8.2	7.5	7.9	7.7	6.3	6.0		
	PrecAvg	18.8	1.4	0.9	0.8	0.8	1.5	3.1	2.8	2.6	1.7	0.9	1.1	1.1		
(15) Precipitation	PrecMax	28.3	4.7	2.7	2.4	1.9	3.7	6.9	6.9	7.3	5.1	2.3	2.3	3.9		
	PrecMin	9.9	0.3	0.1	0.1	0.1	0.1	0.2	0.4	0.5	0.2	0.2	0.0	0.0		
	PrecStd	4.3	1.0	0.6	0.5	0.5	1.0	1.9	1.7	1.7	1.2	0.5	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	42.8	50.2	69.6	80.1	82.1	86.7	85.5	80.6	68.5	52.8	45.9		
		MCWB	37.1	36.2	41.1	50.5	56.7	60.8	65.0	63.8	60.0	53.0	43.1	39.2		
	2%	DB	40.5	39.6	45.3	62.6	74.1	77.8	81.3	80.8	74.9	60.5	44.8	41.1		
		MCWB	35.8	34.5	37.6	46.9	54.4	59.0	62.5	62.9	57.9	48.6	38.8	35.8		
	5%	DB	37.6	36.9	42.0	57.2	70.1	74.3	77.9	76.8	70.1	55.5	40.6	37.0		
		MCWB	33.5	32.5	35.7	44.0	52.4	57.4	61.6	60.5	55.9	45.4	35.7	32.7		
	10%	DB	33.3	33.3	39.1	52.8	66.0	70.8	74.6	73.2	65.2	51.7	37.5	32.6		
		MCWB	30.2	29.8	33.8	41.5	50.5	55.8	60.0	59.1	53.3	43.2	33.7	29.3		
	0.4%	WB	37.8	36.9	41.6	51.6	58.6	63.4	67.5	67.1	61.9	54.3	43.2	39.2		
		MCDB	42.6	41.7	48.4	67.7	75.6	76.7	79.9	79.4	76.4	67.0	50.8	45.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.9	34.8	38.5	47.7	55.7	61.0	65.1	64.3	58.9	49.3	39.6	35.9		
		MCDB	40.2	39.3	44.4	61.3	70.8	73.9	77.1	77.2	72.5	58.4	44.8	40.7		
	5%	WB	33.6	32.7	36.0	44.6	53.5	59.0	63.0	62.2	56.6	46.5	36.1	32.8		
		MCDB	37.3	36.8	41.4	55.6	67.5	71.1	74.7	73.9	68.2	54.0	39.8	36.8		
	10%	WB	30.3	29.8	34.0	42.2	51.4	57.1	61.1	60.2	54.0	43.9	33.8	29.3		
		MCDB	33.4	33.2	39.0	51.7	64.3	68.2	72.0	70.8	63.6	50.6	37.0	32.5		
	5% DB	MDBR	17.2	18.7	18.5	19.6	22.3	20.7	21.1	22.2	22.0	18.4	15.7	16.7		
		MCDBR	18.9	20.1	19.7	28.8	29.9	27.4	28.2	29.4	31.8	25.6	18.5	20.0		
		MCWBR	15.4	16.3	14.2	16.4	14.8	12.9	13.1	13.8	16.7	15.6	13.8	16.5		
		MCWBR	18.6	19.4	19.3	26.4	27.0	24.1	25.3	26.6	30.0	24.6	17.8	20.1		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.253	0.304	0.327	0.347	0.350	0.350	0.350	0.311	0.286	0.247	0.236		
		taud	2.366	2.413	2.326	2.390	2.393	2.401	2.432	2.418	2.534	2.564	2.451	2.271		
	Ebn at Noon		234	272	277	285	284	284	282	274	274	257	230	202		
		Edh at Noon	16	22	30	33	35	35	33	31	24	19	14	13		
(44) All-Sky Solar Radiation	RadAvg		239	531	985	1465	1733	1895	1860	1520	1049	570	262	172		
	RadStd		22	40	75	88	123	154	125	128	99	60	22	14		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page